MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

DISCIPLINE SPECIFIC ELECTIVE (DSE)- MG4DSEGEO201- CLIMATOLOGY

ANSWER SCHEME- 25901292

Part A

Short Answer in 60 words
Answer any **5** questions
Each question carries **2** marks.

1. It refers to the continuous exchange of heat (energy) between the Earth's surface and the atmosphere through processes like radiation, conduction, and convection, maintaining the Earth's energy balance. **Definition- 1mark. Mention of processes- 1 mark.**
2. The Coriolis force is an apparent force caused by the rotation of the Earth, which deflects the path of moving objects such as winds and ocean currents. **Definition- 1 mark.** Northern Hemisphere- winds are deflected to the right. Southern hemisphere- winds are deflected to left- **1 mark.**
3. Helps in understanding and predicting droughts, floods and monsoons. Strongly affects Indian monsoon rainfall and agriculture. **One significance- 2 marks.**
4. The equatorial or tropical fronts are called intertropical fronts (ITF) or intertropical convergence (ITC). These fronts represent the meeting ground of north-east and south-east trade winds. **Definition- 2 marks.**
5. **Sea Breeze (Day)-** Land heats faster. Air rises over land**- 1 mark**. Cool air from sea moves inland. **Land Breeze (Night)-** Land cools faster. Air flows from land to sea**- 1 mark**.
6. Refers to direct deposition of water vapour without hygroscopic nuclei on the surfaces of objects on the ground after condensation of moist air caused by radiation and contact cooling of air lying above the ground surface. **Definition- 2marks.**

Part B

Short Notes in 250 words
Answer any **6** questions
Each question carries **8** marks.

7. **Introduction- 1 mark**. Layers of atmosphere and Temperature pattern- Troposphere, Stratosphere, Mesosphere, Thermosphere- **Detailed explanation- 6 marks. Diagram- 1mark.**
8. **Introduction- 1 mark**. Latitudinal Distribution of Pressure Belts – **Detailed explanation- 5 Marks.** Factors Affecting Pressure Distribution – **Explanation -2 Marks.**
9. Types- **1. Cold anticyclones-** after originating in the arctic regions cold anticyclones advance in easterly and south-easterly directions. Low thickness, smaller, very few higher than 3000m. **2. Warm anticyclones-** originate in the belt of subtropical high pressure where winds diverge in opposite directions. dynamically induced, large in size, sluggish movement, associated with light wind, cloudless sky and clear weather. **3. Blocking anticyclone-** develop due to obstruction in the air circulation in the upper troposphere. Small in size, moves very slowly. **Detailed Explanation- 8 marks.**
10. On the basis of altitude clouds have been classified into high, medium and low clouds. **Introduction- 1 mark.**

    A) High clouds- (height, 3-8 km over the poles, 5-13 in temperate regions, 5-18 km in tropical regions). Includes cirrus clouds, Cirro-cumulus clouds, Cirro- stratus clouds.

    B) Middle clouds- (height 2-4 km over the poles, 2-7 km in the temperate regions, 2-8 km in the tropical regions). Includes Alto-stratus clouds, Alto-cumulus clouds, Nimbo- stratus clouds.

    C) Low clouds- (height, ground surface to 2.5 kilometers). Includes Strato- cumulus clouds, Stratus clouds, cumulus clouds and Cumulo- nimbus clouds.

    Explanation about these types and their characters with figures- 7 marks.
11. Introduction- **1 mark**. Reflection of solar radiation- **3marks.** Concept of Albedo- **3 marks.** Significance- **1mark.**
12. Precession is the wobbling motion of earth's rotational axis, similar to a spinning top. Caused by gravitational forces of the Sun and Moon on Earth's equatorial bulge. Completes one cycle in about 26,000 years. Changes the position of stars over long periods. Influences long-term climate changes (Milankovitch cycles). **Definition- 1 mark. Explanation- 3.5 marks. Significance- 2.5 marks. Diagram- 1 mark.**

13. Polar cell- **Definition- 1.5 mark.** Location and winds, Coriolis effect, polar front, circulation loop- **Explanation- 4.5 marks. Diagram- 2 marks.**
14. Origin of Indian monsoon- Sir Edmund Halley's theory- **Explanation- 4 marks**. Modern Theory: Air mass theory- **Explanation- 4 marks.**

Part C

Essay in 400 words
Answer any **1** question
Each question carries **12** marks.

15. Introduction- **1 mark.** Horizontal Distribution of Temperature- Meaning, Factors Affecting Horizontal Distribution- **4 marks. Example- 1 mark. Diagram- 0.5 mark.** Vertical Distribution of Temperature- **Meaning- 1 mark**. Temperature variation**- 3 marks, Example-1 mark. Diagram- 0.5 mark.**
16. Types of precipitation- Convectional rainfall, Orographic rainfall, Cyclonic or frontal rainfall.

    Convectional rainfall- occurring due to thermal convection currents caused due to insolational heating of ground surface **– 4marks.**

    Orographic rainfall- occuring due to ascent of air forced by mountain barrier- **4 marks.**

    Cyclonic or frontal rainfall- occurring due to upward movement of air caused by convergence of extensive air masses**- 4 marks.**